SEKHUKHUNE DISTRICT HOSPITAL IT INFRASTRUCTURE PROJECT

GROUP 6

Task 2

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Hardware requirements and proposed solutions

A. Network infrastructure hardware

The hospital needs a reliable network infrastructure that connects its wards, offices, clinical departments, servers and authorised users.

Hardware requirement	Proposed solution	Purpose and expected benefit
Core network switches	Two managed 48-port PoE switches	Provide central network connectivity and support VLANs, network monitoring and traffic prioritisation. Two switches can improve resilience when appropriately configured.
Access switches	Managed 24-port and 48-port PoE switches	Connect computers, printers, wireless access points and IP phones across hospital departments.
Fibre-optic equipment	Fibre-optic cables, SFP/SFP+ transceivers and aggregation switches	Establish a high-speed connection between the hospital and the municipal network, as well as between network distribution points.
Structured cabling	Cat6/Cat6A Ethernet cabling, patch panels and patch cables	Provide organised, reliable wired connectivity to wards, offices, workstations and network equipment.
Wireless access points	Business-grade wireless access points	Provide wireless access for authorised staff using laptops, tablets and other approved devices in appropriate areas of the hospital.
Network cabinets and racks	Server racks and distribution cabinets	Securely house and organise network switches, patch panels, servers and related equipment.

B. Server and data-storage hardware

Hardware requirement	Proposed solution	Purpose and expected benefit
Application and virtualisation servers	Two enterprise-grade rack servers	Host supported hospital applications and infrastructure services. Virtualisation allows multiple server workloads to run efficiently on physical hardware.
Backup server	Dedicated backup server	Manage backup jobs and recovery copies of important data.
Network-attached storage (NAS)	Enterprise-grade backup storage with suitable capacity	Store backup data and support recovery after accidental deletion, hardware failure or a cyberattack.
Server racks	Lockable 42U server racks	Protect and organise servers, storage equipment and associated components.
Storage and monitoring components	Compatible enterprise drives and hardware-monitoring components	Support adequate storage capacity and help detect hardware faults before they cause extended downtime.

C. Cybersecurity and network protection hardware

Because the hospital handles confidential patient information, its infrastructure must protect against unauthorised access, malware and network attacks.

Hardware requirement	Proposed solution	Purpose and expected benefit
Firewalls	Two business-grade firewalls configured for high availability	Inspect and control traffic entering, leaving and moving between network segments. A secondary firewall can take over if the primary fails.
Secure network equipment	Managed switches supporting VLANs and access controls	Limit unnecessary communication between departments and help isolate compromised devices.
Security management infrastructure	Compatible monitoring and logging hardware or virtual appliances	Collect network and security events to help the ICT team identify suspicious activity.

Secure server-room equipment	Lockable racks and appropriate physical access controls	Restrict unauthorised physical access to servers, storage and network equipment.

D. End-user computing hardware

The hospital has approximately 600 staff members. Staff need suitable devices to access patient information, communicate with other departments and perform administrative work.

Hardware requirement	**Proposed solution**	**Purpose and expected benefit**
Desktop computers	Business-grade desktop PCs for fixed workstations	Enable authorised staff to access hospital applications, patient records and administrative systems.
Laptops	Business-grade laptops for staff who need mobility	Support work across departments, meetings and approved mobile working activities.
Monitors and peripherals	Monitors, keyboards, mice and docking stations where required	Improve usability and support efficient work at clinical and administrative workstations.
Document scanners	Departmental document scanners	Allow authorised staff to digitise paper-based documents and reduce reliance on physical files.
Biometric readers or smart-card readers	Compatible authentication devices, if required by the chosen access-control system	Provide an additional method of verifying staff identity where justified by security requirements.

E. Power protection and environmental hardware

Network and server equipment must remain protected from power interruptions and unsuitable operating conditions.

Hardware requirement	**Proposed solution**	**Purpose and expected benefit**

Uninterruptible power supplies (UPSs)	Enterprise-grade UPS units for servers and critical network equipment	Provide temporary power during outages and protect equipment from certain power disturbances.
Rack-mounted power distribution units	Suitable rack-mounted power distributors	Distribute power safely and organise connections within equipment racks.
Cooling and ventilation	Appropriate server-room cooling and ventilation equipment	Maintain suitable operating temperatures and reduce the risk of overheating.
Environmental sensors	Temperature, humidity and water-leak sensors where appropriate	Alert the ICT team to environmental conditions that could damage equipment.
Standby generator or backup power integration	Connect critical ICT equipment to an appropriate standby power source where available	Support longer outages that exceed UPS runtime.

System and Application Software Solution for the Sekhukhune District Hospital

Introduction

The proposed system and application software solution is designed specifically for the hospital in the Sekhukhune District. The hospital currently has outdated ICT equipment, separate systems for different activities, concerns about data storage, missing patient files and limited ICT support. Therefore, the software solution helps the hospital manage its existing operations more effectively while working together with the proposed secure and reliable network infrastructure.

System software will provide the basic environment required for the hospital's computers and network devices to operate securely and reliably. Application software will support the hospital's medical operations, finance, supply-chain activities and supplier communication.

1.REQUIREMENTS

System Software Requirements

System software is the software that manages the computers, network resources and other technology used by the hospital.

***Operating system**

The hospital requires a modern and supports operating system for its computers and servers.

The operating system should:

- manage computer hardware and software resources;
- allow hospital employees to run the required applications;
- support network connectivity;
- support user accounts and access permissions;
- provide security features;
- receive security and software updates;
- support reliable storage and retrieval of hospital information.

This is particularly important because the hospital currently has 15 very old computers. The existing computers therefore need to be assessed and, where necessary, replaced or upgraded so that the required software can operate properly.

***Server operating system**

A server operating system should be used for the central management of hospital applications and information.

The server environment should support:

- centralised storage;
- user authentication;
- application services;
- controlled access to hospital information;
- network management;
- data management;
- system monitoring.

Centralising important information helps address the scenario's concern about data storage and missing patient files.

***Database management system**

A database management system is required to store and organise information used by the hospital.

The database should provide controlled storage for information associated with:

- patients;

- medical operations;
- finance;
- supply-chain activities;
- suppliers.

The database should allow authorised employees to search for and retrieve information when required. This addresses the problem of patient files being difficult to locate or missing.

***Network management software**

Network management software should be installed to help the limited ICT team monitor the hospital's network. This is important because the hospital has only three ICT support staff.

The software should allow ICT staff to:

- monitor network availability;
- identify connectivity problems;
- monitor network devices;
- identify failures;
- monitor performance;
- identify possible security problems.

This reduces the amount of manual monitoring required from the three ICT staff.

***Security software**

Security software is required because the is a lack of network-security measures.

The system should include:

- firewall protection;
- anti-malware protection;
- user authentication;
- access control;
- security updates;
- monitoring of unauthorised access.

Only authorised users should be able to access sensitive hospital information.

***Backup software**

Backup software is particularly important because the scenario identifies data-storage concerns and missing patient files. The system should automatically back up important hospital information so that information can be recovered if the original data becomes unavailable. The backup system should cover the information stored by the hospital's main applications.

Application Software Requirements

The application software should support the four main areas already identified which are medical operations, Finance, Supply-chain management, Supplier communication. Rather than creating separate applications that cannot communicate with one another, the proposed solution should provide an integrated hospital information system.

Hospital Information Management Application

The main application should provide a central system through which authorised hospital employees can access the information required for their work. The system should contain separate modules for the hospital's different activities while allowing information to be stored centrally.

***Patient and Medical Operations Section**

This part should allow authorised hospital staff to:

- create patient records;
- retrieve patient records;
- update patient information;
- record relevant medical information;
- search for existing patient files;
- maintain patient information centrally.

The purpose is to reduce the problem of missing patient files.

***Finance**

The finance section should support the hospital's existing financial activities.

It should allow authorised finance employees to:

- capture financial information;

- retrieve financial records;
- update financial information;
- maintain financial records centrally;
- produce relevant financial information when required.

***Supply-Chain Management**

The supply-chain section should support the hospital's supply-chain activities.

It should allow authorised employees to:

- record supply requirements;
- maintain supply records;
- track supply information;
- access supply-chain information;
- communicate relevant supply information to suppliers.

This connects directly with the supply-chain management system already mentioned in the scenario.

***Supplier Communication**

The supplier communication function should provide a structured way for the hospital to communicate with suppliers.

It should allow authorised users to:

- maintain supplier information;
- send supplier-related communication;
- receive supplier communication;
- store relevant communication;
- retrieve previous communication when required.

This reduces the reliance on disconnected communication and helps keep supplier information available within the hospital system.

2.SOLUTIONS

User Management and Access Control

Because the hospital has approximately 600 staff, the application will have controlled user management. Different employees should receive access according to their responsibilities.

Centralised Patient File System

One of the most important application requirements is the management of patient files. There are missing patient files as a problem. Therefore, the proposed application is going to replace reliance on scattered or difficult-to-locate files with a centralised electronic patient record system.

The process would be:

1. Patient registered
2. Patient record created
3. Medical information captured
4. Record stored centrally
5. Authorised staff retrieve record when required

This means authorised staff in the relevant wards and offices can retrieve the patient's information through the hospital network. The system allows records to be searched rather than requiring staff to manually locate a physical file.

Data Storage Solution

The application and system software work together with a central data-storage environment.

The proposed structure is:

1. Hospital computers
2. Secure network
3. Central server/database
4. Applications

The central database stores the information required by the hospital applications. This provides a single controlled location for important information rather than allowing important records to

remain scattered across individual computers. The solution also includes automated backups so that important information can be recovered.

Integration of the Existing Hospital Systems

The hospital currently has separate systems for:

- medical operations;
- finance;
- supply-chain management;
- supplier communication.

The proposed application solution should therefore focus on integration.

Instead of:

Medical system, Finance system, Supply-chain system, Supplier system

The proposed architecture becomes: Integrated Hospital Information System with the four functions operating as separate sections connected to central information management. This allows the hospital to maintain the functions it already needs while improving how the systems work together.

Security Requirements

Security must be included in both system software and application software because the scenario specifically identifies missing network-security measures.

The solution should include:

Authentication, every authorised employee should have an individual username and password.

Authorisation, users should only access the information required for their responsibilities.

Access control, the system should prevent unauthorised users from accessing hospital information.

Firewall protection, the network should be protected against unauthorised network access.

Anti-malware protection, hospital computers and servers should have protection against malicious software.

Software updates, operating systems and applications should be kept updated to reduce security weaknesses.

Backup, important information should be backed up so that it can be restored when required.

Software Requirements and Solutions Summary

REQUIREMENT	SOLUTION
Old computers	Replace computers so they can support the required software
Data storage concerns	Centralised database and storage
Missing Patient files	Electronic central patient records
Separate Hospital systems	Integrated hospital information systems
Lack of security	Authentication, access control, firewall and anti-malware
Limited ICT staff	Network/system monitoring and centralized administration
Data availability	Automated backup and recovery
Network Connectivity	Applications operate through the proposed hospital network
Future growth	Scalable system and application architecture

Implementation of the Software Solution

The software implementation takes place as part of the one-year implementation period

Phase 1: Assessment

The hospital's existing:

- 15 computers;
- 3 printers;
- existing systems;
- data;
- network environment should be assessed.

The purpose is to determine what can be retained and what needs to be upgraded or replaced.

Phase 2: System software installation

The required operating systems, server software, database management system, security software, backup software and network management software is installed.

Phase 3: Application development/configuration

The integrated hospital application is configured around the hospital's four existing operational areas:

Medical → Finance → Supply Chain → Supplier Communication

Phase 4: Data migration

Existing relevant information is then transferred into the central system so that authorised staff can retrieve the information through the new application. Attention should be given to patient information because the scenario identifies missing patient files as a problem.

Phase 5: User accounts

Accounts should be created for hospital employees according to their responsibilities.

The system should assign appropriate permissions to medical, finance, supply-chain, supplier communication, management and ICT users.

Phase 6: Testing

The system is tested to ensure that:

- users can log in;
- authorised users can access the correct information;
- unauthorised users are restricted;
- patient records can be retrieved;
- finance functions work;
- supply-chain functions work;
- supplier communication works;
- data is stored correctly;
- backups work;
- the applications operate across the hospital network.

Phase 7: Staff training

Hospital employees are trained to use the new applications. This is especially important because the hospital has approximately 600 staff and the solution changes how information is accessed and managed.

Phase 8: Deployment

After testing and training, the integrated system is deployed across the hospital's wards and offices. The software operates through the proposed secure network connecting the hospital to the municipal network.

The system software provides the foundation for operating, securing and managing the hospital's technology, while the application software provides the functions needed by the hospital's medical, finance, supply-chain and supplier-communication activities.

Most importantly, the solution addresses the major problems: outdated ICT equipment, data-storage concerns, missing patient files, separate systems, lack of security measures and limited ICT support. It also supports the required outcome of a fast, reliable, secure, sustainable and scalable hospital technology environment.

IT Policies and Standard Operating Procedures for the Sekhukhune District Hospital.

1. Network Security Policy

The hospital network should be secure and reliable and accessible only to authorised personnel. To help safeguard hospital systems and patient data, firewalls, endpoint protection, network monitoring and network segmentation should all be put in place.

SOP:

- Set up authorized firewalls and network security mechanisms.
- Keep an eye on the network and system.
- Monitor and investigate suspicious or unauthorized behavior.
- Report security incidents to ICT Department.
- Resolve and record the situation.

2. Access Control Policy

Information and systems in the hospital are available only to authorized employees. Employees should each have their own account and access to it should be dependent on their role. Wherever possible, Active Directory and MFA should be used.

SOP:

- Accept and approve employee request for access.
- Make a personal user account.
- Set the permissions that are needed.
- Set up MFA as necessary.

- Test access of the user.
- Update or eliminate access for an employee when he or she changes roles or exits the hospital.

3. Implement a Data Backup and Disaster Recovery Policy:

Regular backup of hospital data is essential due to data loss, security threats and system failures. Backups need to be securely stored and tested.

SOP:

- Make a list of all critical systems and information to be backed up.
- Follow the approved backup system for scheduled backups.
- Ensure that backups have been successfully created.
- Analyse and fix unsuccessful backups.
- Regularly test for restoration.
- Document backup/recovery outcomes.

As for the proposed infrastructure the project will feature a dedicated backup server, backup storage and DR as part of that.

4. IT Change Management Policy:

All major changes in the hospital's IT infrastructure are to be approved, planned and documented. Any changes need to be implemented in a manner that will cause the least disruption to healthcare services.

SOP:

- Submit a change request, stating the required change.
- Identify the risks and potential impact on the hospital.
- Get the responsible authority's permission.
- Plan the change for the right time.
- Institute and pilot the change.
- Verify correct operation of affected systems.
- Record the final results.

This is essential as the infrastructure upgrade is being carried out, while the hospital services are still operational.

5. IT equipment and Asset management policy:

All hospital IT equipment should be authorized, recorded, secured and maintained. Authorized ICT staff only can install, move or change critical network equipment.

SOP:

- Get and review new equipment.
- Document equipment and serial number.
- Designate and place equipment.
- Install and set-up the equipment.
- Test the equipment prior to use.
- Update hospital asset register.

6. Incident management and IT Support policy:

Any IT related issues should be reported to and logged in the ICT Department. Any incident that relates to patient care or is serious to the hospital's operation should be addressed immediately.

SOP:

- Report IT issue to ICT.
- Take a note and document the incident.
- Identify what it is and its significance.
- Troubleshoot and resolve the problem.
- Raise serious and unresolved incidents as appropriate.
- Test affected system following the resolution.
- Close and document the incident.

Limited ICT staffing is identified as a risk factor so it is crucial to have appropriate documentation, training and escalation.

7. Staff Training and Awareness Policy:

The staff of the hospital should be given suitable training related to the new IT systems, cyber security and correct IT procedures. Further technical training is required for ICT staff as needed.

SOP:

- Determine staff training needs.
- Prepare training material.
- Conduct training sessions.
- Demonstrate correct system usage and security practices.
- Record staff attendance.
- Provide additional assistance where required.

The project includes staff training to facilitate the successful implementation and adoption of the new infrastructure.

8. Policy Review and Compliance:

These policies and SOPs have to be kept current and relevant in the Hospital ICT environment.

SOP:

- Regularly review policies and after significant system changes.
- Identify areas requiring improvement.
- Revise pertinent policy or procedure.
- Obtain management approval.
- Inform involved personnel of the changes.
- Give further training if required.
- Have the current approved version as the official document.

The key project goal of reliable connectivity, 99.5% network performance, secure access, enhanced cybersecurity, minimized downtime and future expansion is supported by these policies.

Network setup

Overview and design goals

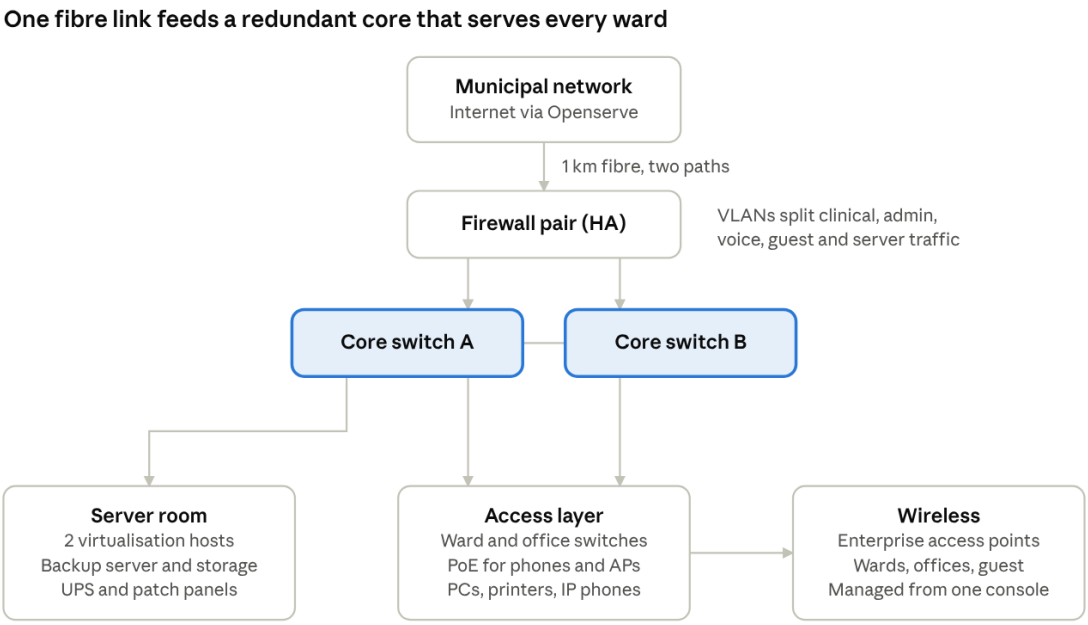
The hospital joins the municipal network over a 1 km dual path fibre link, with a redundant core, segmented VLANs and a secure server room at its center. It serves about 600 staff, 20 wards, 5 offices and roughly 100 patients a day.

Design goals:

- 99.5% network availability, with no single point of failure at the core or the uplink
- Separate clinical, administrative, server, voice, guest and management traffic
- Patient data protected by firewall, MFA, endpoint security and tested backups
- Capacity to add users, wards and devices without rebuilding the network
- Simple enough for a small ICT team (3 staff per slide 4) to run, with monitoring and documentation

Network topology

A three-layer design: a firewall at the edge, two core switches for redundancy, and access switches in every ward and office.



hospital network topology · 4 layers, redundant core

The fibre extends the municipal network about 1 km and ends in the hospital server room. Each access switch has an uplink to both cores, so one failed switch or cable does not take a ward offline. Cabling is Cat6 to desks and fibre between cabinets.

Hardware

Quantities marked "survey" are planning figures to confirm in the site survey; the rest come from the group's slides.

Item	Specification	Qty	Purpose
Fibre extension	Single-mode fibre, about 1 km, two physical paths	1 link	Municipal network to hospital server room
Edge firewalls	Next-generation firewall, high-availability pair	2 links	Filtering, intrusion prevention, VPN
Core switches	Layer 3, 10 GbE uplinks, stacked or paired	2 links	Routing between VLANs, redundancy
Access switches	24/48-port PoE+ gigabit	About 30 (survey)	Wards, offices, phones, access points
Wireless access points	Enterprise Wi-Fi 6 with central management	About 50 (survey)	Wi-Fi in wards, offices and guest areas
Virtualisation servers	Dell PowerEdge-class rack servers	2	Hosts for AD, file, print and application servers
Backup server and storage	Backup server with disk storage, plus off-site or cloud copy	1	Backups and disaster recovery
Structured cabling	Cat6 to desks, fibre between cabinets, patch panels, certification	Whole site	Wired network to every ward and office
Network cabinets	Wall or floor racks with cooling	Per ward block (survey)	House switches and patch panels
UPS units	Rack UPS for server room and cabinets	Per cabinet (survey)	Ride through power cuts and load shedding

Item	Specification	Qty	Purpose
Workstations	Business PCs replacing the 15 old computers	Per needs analysis	Clinical and admin work
Printers	HP LaserJet Enterprise multifunction	8	Replace the 3 existing printers
IP phones	PoE desk phones on the voice VLAN	Per needs analysis	Communication between departments

Software

Layer	Software	Purpose
Server operating system	Windows Server	Runs domain, file, print and application servers
Virtualisation	VMware vSphere or Microsoft Hyper-V	Runs several servers on the 2 hosts with failover
Identity	Active Directory, with role-based accounts	One account per employee, access by role
Multi-factor authentication	MFA on remote, admin and Microsoft 365 sign-ins	Stops stolen passwords being enough
Productivity and email	Microsoft 365 Enterprise E3 (US$39 per user per month, yearly plan)	Email, documents, Teams
Endpoint security	EDR/antivirus licenses for 600 endpoints	Detects malware on PCs and servers
Network security	Firewall intrusion prevention, web filtering, VPN	Blocks attacks and controls supplier access
Backup and recovery	Backup software for VMs, files and databases, with restore testing	Protects patient and finance data
Network monitoring	SNMP monitoring such as Zabbix or PRTG, with alerts	Shows faults early; helps meet 99.5% availability
Device management	Group Policy and patching (WSUS or Intune)	Keeps PCs updated and consistent
Existing business applications	Operations, finance and supply-chain systems already in use	Kept as they are; hosted on the new servers

The project does not build a new hospital information system or healthcare apps (slide 11), it provides the network and servers they run on.

VLANs and IP addressing

Eight VLANs keep patient systems apart from guests and suppliers. The 10.20.0.0/16 range is a proposed example; confirm it with the municipality so it does not clash with their ranges.

VLAN	Name	Subnet	Who or what is on it	Access rule
10	Management	10.20.10.0/24	Switch, AP and firewall admin	ICT staff only
20	Servers	10.20.20.0/24	Virtualization hosts, AD, backup	Reached by staff VLANs on needed ports only
30	Clinical	10.20.30.0/23	Ward PCs and clinical devices	Servers, internet through filter
40	Administration	10.20.40.0/23	Offices, finance, supply chain	Servers, internet through filter
50	Voice	10.20.50.0/24	IP phones	Phone system only
60	Printers	10.20.60.0/24	Network printers	Print server only
70	Guest Wi-Fi	10.20.70.0/23	Visitors and patients' devices	Internet only
80	Supplier access	10.20.80.0/24	Supplier portal access	Procurement systems only, VPN and MFA

Security design

Security is layered so that one failure does not expose patient records.

- **Perimeter:** a firewall pair with intrusion prevention and web filtering; only needed ports open
- **Segmentation:** VLANs with firewall rules between them; guests and suppliers cannot reach clinical systems
- **Identity:** one named account per employee, role-based access, MFA for remote, admin and Microsoft 365 access; accounts removed when staff leave

- **Endpoints:** EDR on all 600 endpoints, automatic patching, encrypted laptops and removable media
- **Data:** encryption of patient data in transit (TLS, VPN) and at rest on servers and backups
- **Backup and recovery:** daily backups to a separate backup server plus an off-site or cloud copy, with restores tested regularly
- **Physical:** locked server room and cabinets, UPS protection, access log
- **Monitoring:** firewall and switch logs reviewed by ICT, with alerts for failed logins and device faults
- **Compliance:** handling of patient information follows POPIA; the policies and SOPs on slides 33 to 42 apply

Rollout and cost

The build fits the one-year deadline in six phases, with the hospital kept running throughout:

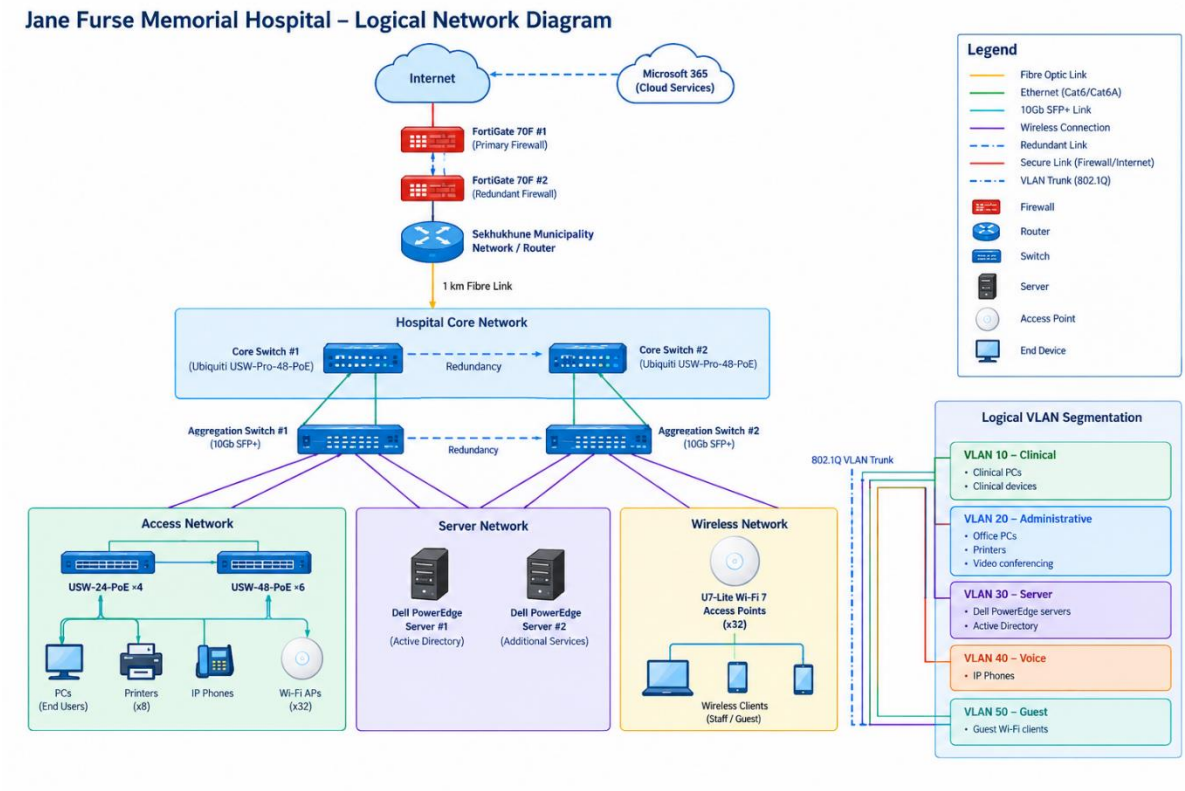
1. Months 1-2: site survey, needs analysis, final design, supplier quotes
2. Months 3-4: fibre extension, cabinets, structured cabling and certification
3. Months 5-6: server room, servers, storage, UPS, backup system
4. Months 7-8: core and access switches, Wi-Fi, firewalls, VLANs
5. Months 9-10: PCs, printers, phones, Active Directory, MFA, endpoint security; staff migrated ward by ward
6. Months 11-12: testing, restore and failover tests, ICT and staff training, handover and documentation

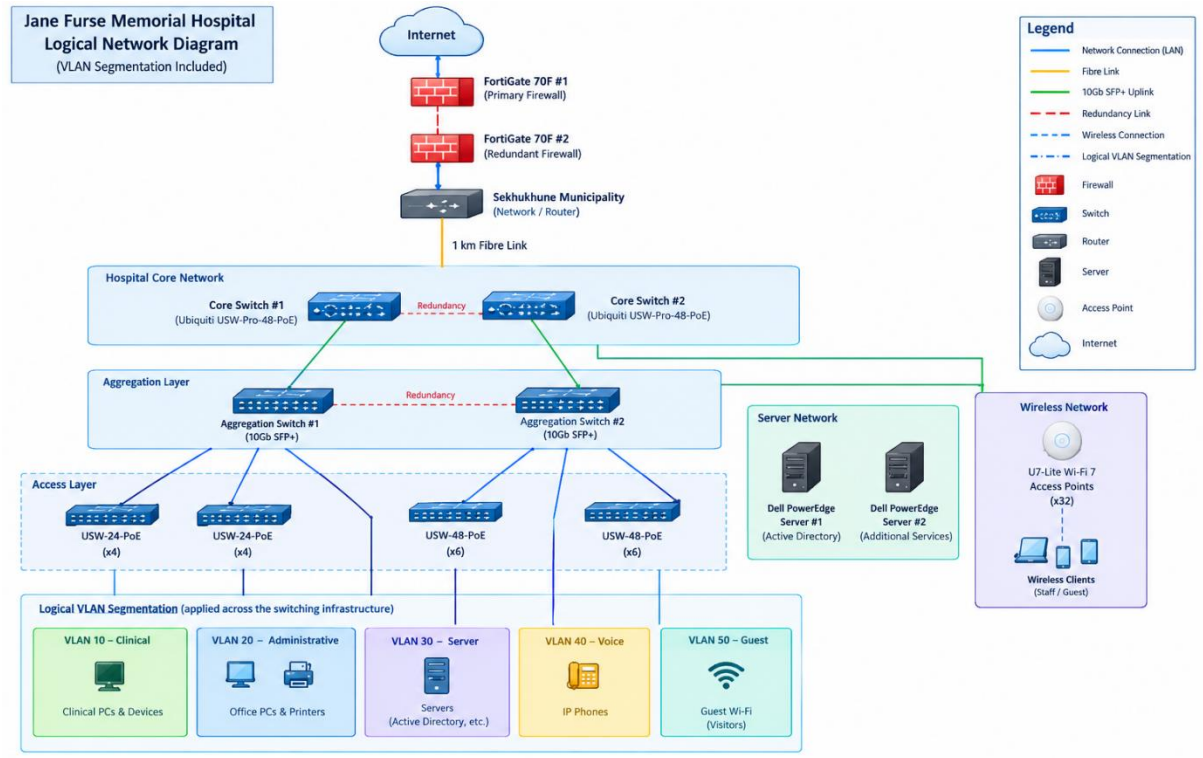
The estimate is R1,975,000. Known major items:

Item	Cost (R)
Endpoint security for 600 endpoints	300,000
Structured cable installation	240,000
Virtualization servers	182,988
Municipal-to-hospital fibre extension	180,000
Cabling certification and testing	150,000
Everything else (switches, PCs, access points, firewalls, backup storage, cabinets, UPS, printers, training)	922,012
Total	**1,975,000**

Detailed Network Architecture Diagrams

Logical Diagram





1. The 1 km Fibre-Optic Link

A one-kilometer fiber-optic link between Jane Furse Memorial Hospital and Sekhukhune Municipality is part of the network design. The hospital's internal network and the municipality's network infrastructure are connected via this interconnection. Because fiber-optic cabling is less prone to electromagnetic interference than copper cabling and allows high-speed data transmission over longer distances, it is appropriate for this connection. Data can be shared using the hospital's core switches, aggregation switches, and access switches to facilitate communication between the hospital's wards, administrative offices, servers, and other connected equipment once the connection has reached the network core.

2. Network Redundancy: FortiGate Firewalls and Core Switches

Redundancy is incorporated into the network design to increase availability and lower the chance of service outages. As long as failover is properly configured, the two FortiGate 70F firewalls that are included provide a primary and a redundant firewall structure, assisting in maintaining network security in the event that the primary firewall fails. To lessen dependency on a single core switching device, two Ubiquiti USW-Pro-48-PoE core switches are also supplied. The hospital's network and aggregation layer are connected centrally by the core switches. When a component or connection fails, traffic can choose a different accessible path thanks to redundant links and suitable backup arrangements. Because employees in a hospital setting rely on dependable access to clinical systems, administrative apps, and shared network resources, this is crucial. The final configuration and testing must support the desired amount of redundancy.

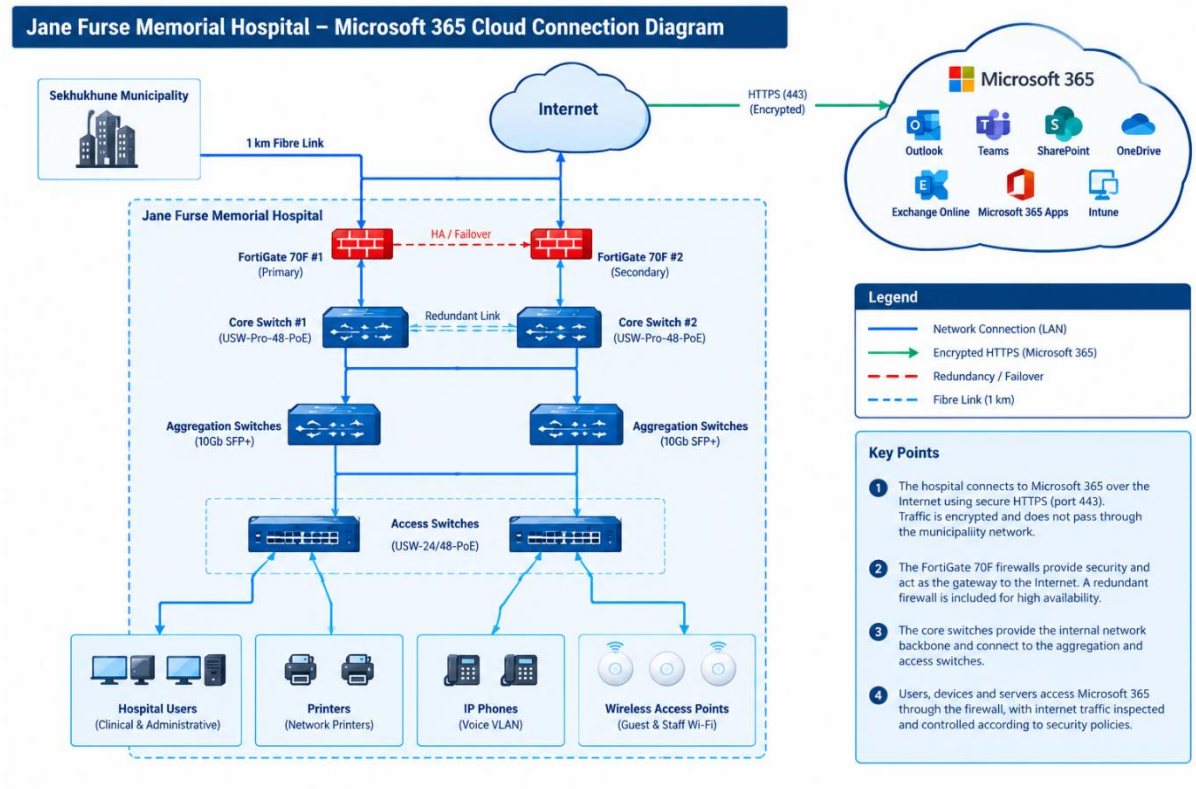
3. VLAN Segmentation

Five Virtual Local Area Networks (VLANs) are included in the logical network diagram to divide network traffic based on its intended use. Clinical PCs and devices are assigned VLAN 10, administrative computers and printers are assigned VLAN 20, servers and Active Directory are assigned VLAN 30, IP phones are assigned VLAN 40, and guest Wi-Fi users are assigned VLAN 50. This division facilitates the implementation of suitable security policies, reduces pointless communication between various groups of devices, and aids in the organization of network traffic. For instance, healthcare systems and internal servers shouldn't be freely accessible to visitors using Wi-Fi. VLAN trunk links transport traffic for several VLANs between compatible network devices, and the VLANs are set up on the network switches. Routing and access-control rules should decide what traffic is allowed when connectivity between VLANs is necessary.

4. Physical Network Diagram

The intended equipment and cabling arrangement within the hospital infrastructure is shown in the physical network diagram. The servers, aggregation switches, firewalls, and core switches are all located in the main ICT room. The access switches in the hospital's distribution cabinets are connected to the aggregation layer. These access switches use the proper cabling to link wireless access points, IP phones, printers, and end-user devices. When high-speed uplinks are needed, fiber-optic cables are utilized, while connections to adjacent devices are supported via Cat6 or Cat6A copper cabling. The physical layout facilitates installation, troubleshooting, and future maintenance by offering an informative picture of equipment placement and cabling paths across the hospital's offices and wards.

Cloud Diagram



The cloud diagram shows how Jane Furse Memorial Hospital uses the internet to link its internal network to Microsoft 365. Through their PCs and other authorized devices linked to the access, aggregation, and core switches, hospital users can access cloud services. Before reaching Microsoft 365, network traffic must first transit through the FortiGate firewalls, which adds another degree of security to external connections. Outlook, Teams, SharePoint, OneDrive, Word, and other Microsoft 365 services facilitate email correspondence, online meetings, document sharing, and departmental cooperation in hospitals. While security features like multifactor authentication and access controls help prevent unauthorized access, connections employ encryption to secure data while it is in transit. By eliminating the requirement to host key productivity services on nearby hospital servers, this cloud-based approach enhances accessibility and cooperation. However, in order to preserve access and safeguard sensitive healthcare information in compliance with POPIA, dependable internet connectivity and proper data-protection measures are required.

Outsourced Services

The hospital has limited ICT staffing capacity; therefore, it could outsource a number of IT services to specialized outside providers. The project charter notes that the hospital presently employs just one ICT support staff member and expressly names a network vendor and Internet service provider as support resources.

Possible outsourced IT services

Outsourced service	What the external provider would do	Why it makes sense for the hospital
Internet/fibre connectivity	Provide and maintain the hospital's fibre connection and internet service	The hospital needs reliable connectivity between the municipality and hospital.
Network maintenance & support	Monitor switches, routers, Wi-Fi, firewalls and network performance	Reduces pressure on the hospital's limited ICT staff.
Cybersecurity monitoring (MSSP)	24/7 monitoring for malware, suspicious activity, intrusion attempts and security incidents	The hospital handles sensitive patient information and faces cybersecurity threats.
Cloud/Microsoft 365 management	Manage Microsoft 365 accounts, email, collaboration and cloud services	Provides specialist support without requiring the hospital to employ additional specialists.
Cloud backup/disaster recovery	Maintain off-site backups and assist with data recovery after a failure or cyberattack	Protects against data loss and supports business continuity.
Hardware maintenance	Repair/replace computers, printers, switches and other equipment	Helps deal with outdated equipment and reduces downtime.
Server management	Monitor and maintain physical/virtual servers, operating systems and storage	Useful where internal ICT staff lack sufficient server expertise.
Network installation	Install fibre, structured cabling, access points and network equipment	Specialist installation can ensure the infrastructure is correctly implemented.
IT help desk	Provide first-line technical support to hospital employees	Particularly useful because the scenario identifies limited ICT support capacity.
Website hosting & maintenance	Host and maintain the hospital/municipality website	Could be outsourced if the hospital needs a public-facing website but does not have internal web-development expertise.
Staff cybersecurity training	Provide security-awareness and phishing training	Helps reduce human-related security risks.
Equipment monitoring	Monitor network/server equipment and alert the hospital to failures	Can help achieve the project's goal of reducing downtime.

As much as these outsourced services are important, there are some which should be prioritised:

1. Internet/Fibre Service Provider
The hospital needs reliable connectivity, and the project specifically involves extending the municipal fibre connection to the hospital.

2. Managed Network Support
Due to the hospital's low ICT support capability, the network might be maintained and monitored by an outside network specialist.

3. Managed Cybersecurity / SOC service
This would be particularly valuable because the hospital handles sensitive patient information and the project identifies phishing, spyware, ransomware and unauthorised access as risks.

4. Cloud Backup / Disaster Recovery
An external provider could maintain secure off-site backups and assist with restoring systems after a major failure or cyberattack.

5. Microsoft 365 Support
Since Microsoft 365 is part of the proposed infrastructure, the hospital could outsource administration, configuration and technical support rather than requiring its small ICT team to manage everything themselves.

6. Hardware & Network Maintenance
A vendor could provide preventative maintenance and on-site support for critical equipment.

By outsourcing these services, the hospital's internal ICT team could concentrate on vital internal systems and user support while gaining access to specialized technical expertise. However, since the hospital manages sensitive patient data and must make sure that third-party access conforms with security and privacy regulations, outsourced providers should be carefully chosen. The Internet service provider and network vendor are both listed as external support resources in the project charter.